8.2 Scalars and Vectors

Definition 8.1

A scalar is a quantity that is determined by its magnitude.

For instance, distance, length, speed, temperature, voltage, mass, pressure, and work are scalars.

Definition 8.2

A **vector** is a quantity that is determined by both its magnitude and its direction and hence it is a directed line segment.

For instance, force, displacement, and velocity (which gives the speed and direction of the motion) are vectors.

We denote vectors by lower case letters with arrow. A two dimensional vector is a directed line segment in $\mathbb{R}^2$ and a three dimensional vector is a directed line segment in $\mathbb{R}^3$.

8.3 Representation of a vector and types of vectors

A vector has a tail and a tip. Consider the diagram as in Fig. 8.1.

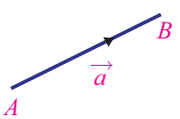


Fig 8.1

Definition 8.3

The tail point A is called the **initial point** and the tip point B is called the **terminal point** of the vector $\vec{a}$. The initial point of a vector is also taken as origin of the vector.

The initial point A of the vector $\vec{a}$ is the original position of a point and the terminal point B is its position after the translation.

The length or magnitude of the vector $\vec{a}$ is the length of the line segment AB and is denoted by $|\vec{a}|$.

The undirected line AB is called the **support of the vector** $\vec{a}$.

To distinguish between an ordinary line segment without a direction and a line segment representing a vector, we make an arrow mark for the vector as $\overrightarrow{AB}$ and $\vec{a}$. So AB denotes the line segment.

Definition 8.4

If we have a liberty to choose the origin of the vector at any point then it is said to be **free vector,** whereas if it is restricted to a certain specified point then the vector is said to be **localized vector.**

Upto vector product we will be dealing with free vectors only. Localised vectors are involved in finding equations of straight lines.

Definition 8.5

Co-initial vectors are having the same initial point. On the other hand, the **co-terminous vectors** are having the same terminal point.

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